1. recall that ores are rocks that contain varying amounts of minerals from which metals can be extracted;
2. recall that for some minerals, large amounts of ore need to be mined to recover small percentages of valuable minerals (for example in copper mining);
3. recall examples of metals that can be extracted by heating the oxide with carbon (for example zinc, iron and copper (technical details not required));
4. recall that when a metal oxide loses oxygen it is reduced while the carbon gains oxygen and is oxidised;
5. understand that some metals are so reactive that their oxides cannot be reduced by carbon;
6. be able to balance unbalanced symbol equations;
7. recall and use state symbols: (s), (l), (g) and (aq) in equations;
8. be able to use the Periodic Table to obtain the relative atomic masses of elements;
9. **be able to calculate the mass of the metal that can be extracted from a mineral given its formula or an equation;**
10. describe electrolysis as the decomposition of an electrolyte with an electric current;
11. understand that electrolytes include molten ionic compounds;
12. describe what happens to the ions when an ionic crystal melts;
13. recall that, during electrolysis, metals form at the negative electrode and non-metals form at the positive electrode;
14. describe the extraction of aluminium from aluminium oxide by electrolysis;
15. **show that during electrolysis of molten aluminium oxide the positively charged aluminium ions gain electrons from the negative electrode to become neutral atoms;**
16. **show that during electrolysis of molten aluminium oxide, negatively charged oxide ions lose electrons to the positive electrode to become neutral atoms which then combine to form oxygen molecules;**
17. **use ionic theory to explain the changes taking place during the electrolysis of a molten salt (limited to using diagrams or symbol equations to account for the conductivity of the molten salt and the changes at the electrodes);**
18. recall the properties of metals related to their uses (limited to strength, malleability, melting point and electrical conductivity);
19. explain the properties of metals in terms of a giant structure of atoms held together by strong metallic bonding;
20. **understand that in a metal crystal there are positively charged ions held closely together by a sea of electrons that are free to move;**
21. evaluate, given appropriate information, the impacts on the environment that can arise from the extraction, use and disposal of metals.

Synthesis provides many of the chemicals that people need for food processing, health care, cleaning and decorating, modern sporting materials and many other products. The chemical industry today is developing new processes for manufacturing these chemicals more efficiently and with less impact on the environment.

In this context, the module explores related questions which chemists have to answer: ‘How much?’ and ‘How fast?’ in the context of the chemical industry. Quantitative work includes the calculation of yields from chemical equations and the measurement of rates of reaction.

A further development of ionic theory shows how chemists use this theory to account for the characteristic behaviours of acids and alkalis.

Topics
C6.1 Chemicals and why we need them
The scale and importance of the chemical industry. Acids, alkalis and their reactions. Neutralisation explained in terms of ions.
C6.2 Planning, carrying out and controlling chemical synthesis
Planning chemical syntheses. Procedures for making pure inorganic products safely. Comparing alternative routes to the same product. Calculating reacting quantities and yields. Measuring purity by simple titration. Controlling the rate of change.

ICT Opportunities

This module offers opportunities for illustrating the use of ICT in science. For example, logging data, storing it and displaying it in a variety of formats for analysis.

Use of ICT in teaching and learning can include:

- video clips to illustrate the manufacture of chemicals on a large-scale in industry;
- sensors and data loggers to monitor neutralisation reactions and the rates of chemical changes.